

Input = 1

Find the new weight

$$\text{New weight} = \text{old weight} + \eta + \text{Error} \times \text{Input}$$

$$= 0.5 + (0.2)(0.4)(1)$$

$$= 0.5 + 0.08$$

$$= 0.58$$

Linear Algebra:

1. scalar is a single numerical value has only magnitude and no direction.

2. vector

→ A vector is an ordered collection of numbers arranged in a single row or single column.

vector addition

$$\begin{bmatrix} 2 & 4 & 6 \end{bmatrix} + \begin{bmatrix} 1 & 3 & 5 \end{bmatrix} = \begin{bmatrix} 3 & 7 & 11 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 3 \end{bmatrix} \times \begin{bmatrix} 3 \end{bmatrix} = \begin{bmatrix} 3 & 6 & 9 \end{bmatrix}$$

Matrix:

Rectangular arrangement of numbers organized into rows & columns.

Matrix Addition:

$$A = \begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix} \quad B = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

$$A + B = \begin{bmatrix} 3 & 5 \\ 7 & 9 \end{bmatrix}$$

Matrix subtraction:

$$A - B = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

Matrix Multiplication:

Process of multiplying rows of first matrix with

$$X = \begin{bmatrix} 2 & 3 \end{bmatrix}$$

$$W = \begin{bmatrix} 4 \\ 5 \end{bmatrix}$$

$$Y = XW$$

Machine learning problem based on linear algebra.

Vector Addition:

Two students have feature vectors.

$$\text{Salary} = (8 \times 6) + 5 = 48 + 5 = 53$$

one
Input
~~2+3~~ weight
Bias

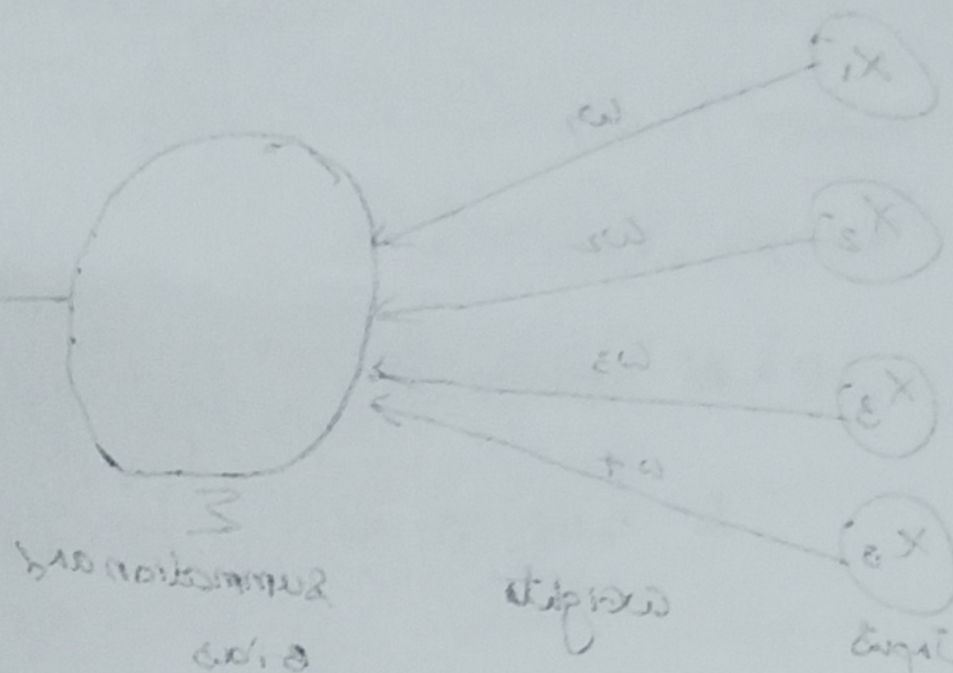
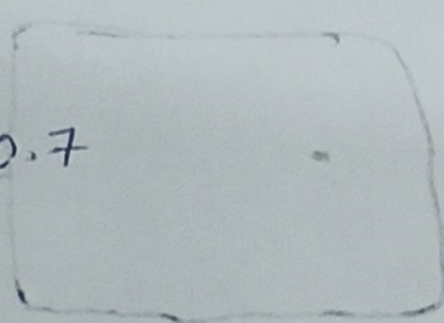
Learning Rate = 0.5

Activation linear

$$\text{Next} = (3)(0.2) + 0.1 = 0.7$$

$$\text{output} = 0.7$$

$$\text{Error} = 1 - 0.7 = 0.3$$



Gradient descent

optimization algorithm that minimize the error by updating the weight.

1. Batch Gradient Descent: use all training the weight.

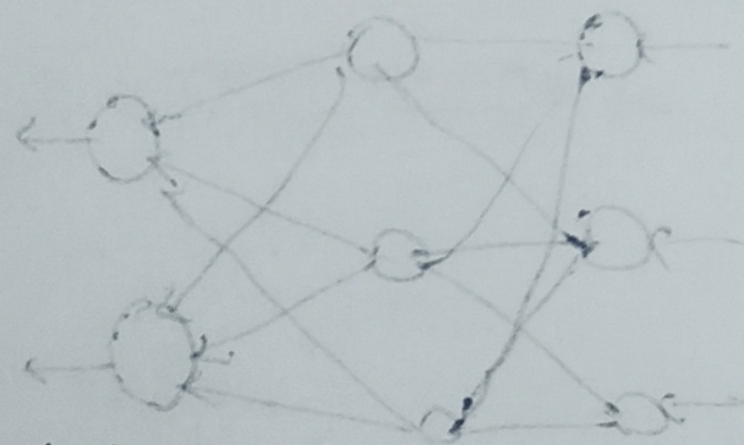
Advantages:

disadvantages:

Working of SGD

Ex	Target
x_1	19
x_2	15
x_3	20
x_4	25
x_5	30

Training sequence
input $x_i \rightarrow$ update



2. mini Batch Gradient Descent:

Instead of using one sample or the entire dataset (MBGD) uses a small batch of samples.

12/11/2024

Curse of Dimensionality

The Curse of Dimensionality refers to the challenge. It becomes more difficult for the model to identify meaningful patterns.

How to overcome the Curse of Dimensionality

1. Feature selection: Choose only the most relevant
2. Use appropriate Algorithms:
3. Dimensionality Reduction.

Probability is the measure of how likely an event is to occur. The probability of an event lies between 0 and 1

$$P(E) = \frac{\text{No. of Favorable outcomes}}{\text{No. of possible outcomes}}$$

* Complement Rule $P(A') = 1 - P(A) = 1 - 0.30 = 0.70$

Random variable

A Random variable is a variable whose value depends on the outcomes of a random experiment.

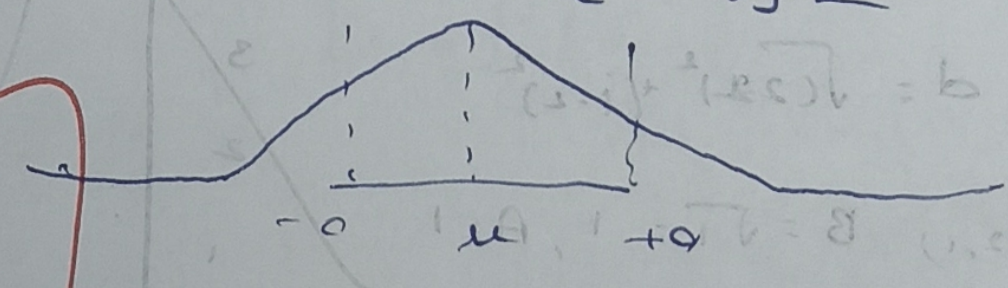
- a) discrete
- b) continuous

4. probability Distribution, 5. Mean = $\frac{\sum x}{n} = \frac{79+75+80+85+90}{5} = 80$

- Bernoulli
- Binomial
- Normal
- Poisson

6. Variance:

$$\text{Var}(X) = E(X^2) - [E(X)]^2$$



7. Standard Deviation

$\sigma = \sqrt{\text{Variance}}$ variance → Feature selection & measuring data spread.

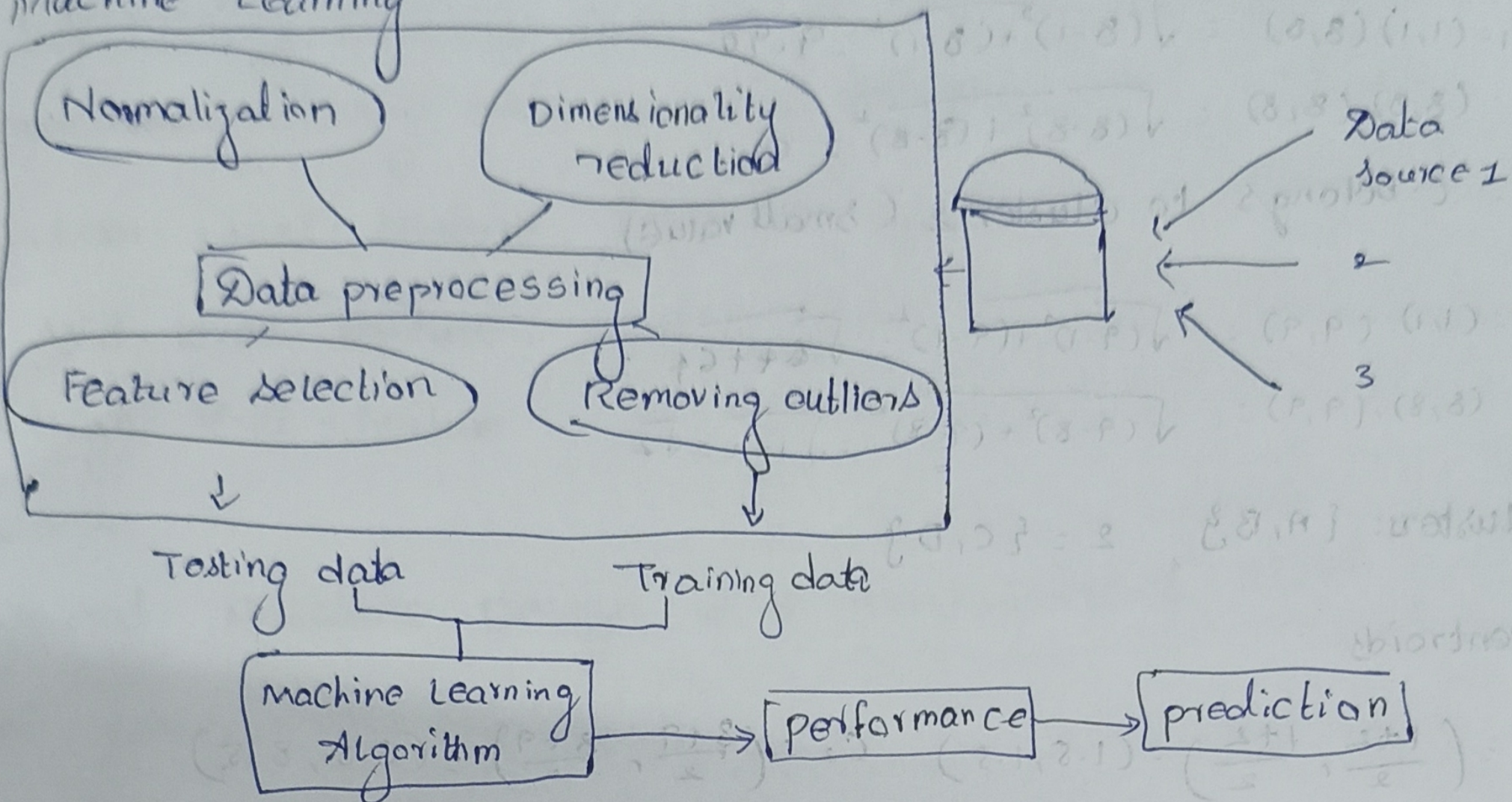
Variance = 2.67, Standard Deviation = $\sqrt{2.67} = 1.63$

Mark (X)	Mean (μ)	(X - μ)	(X - μ) ²
79	80	-10	100
75	80	-5	25
80	80	0	0
85	80	5	25
90	80	10	100

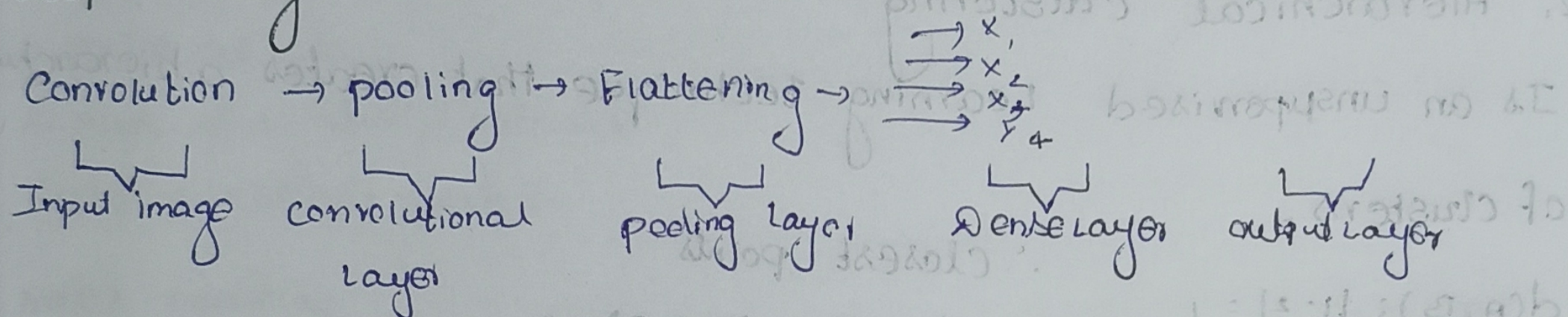
$$\sigma^2 = \frac{250}{5} = \sigma^2 = 50$$

$$\sigma = \sqrt{\text{variance}} = \sigma = \sqrt{50} = 7.07 = \text{standard Deviation}$$

Machine Learning



Deep learning



Feature map

A Feature Map is the output generated by applying a Convolutional filter to an input image or a previous layer within a CNN.

$$I = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

Kernel 2x2

$$K = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 5 & 6 \\ 8 & 9 \end{bmatrix}$$

$$= (5 \times 1) + (6 \times 0)$$

$$\begin{bmatrix} 3 & 2 & 1 \\ 1 & 0 & 2 \\ 2 & 1 & 3 \end{bmatrix}$$

Kernel

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 3 & 1 \\ 2 & 3 \end{bmatrix}$$

$$3(-2) + 2(3+4) + 1(1-6) = -6 + 14 + 1 = -6 + 15 = 9$$

Unsupervised Learning: Is a type of Machine Learning in which the algorithm learns patterns.

1-K-Mean clustering

K → Number of clusters

Cluster → Group of similar data points

Distances

Centroid

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$A(1,1), (2,2), (8,8), (9,9)$$

$$K=2, C_1(1,1), C_2(8,8)$$

$$d(A, C_1) = \sqrt{(1-1)^2 + (1-1)^2} = 0$$

$$C_2 = \sqrt{(8-1)^2 + (8-1)^2}$$

$$= \sqrt{98} = 9.90$$

Representing Learning: is a machine learning technique in which a model automatically means the useful features.

1. Low-level Feature: First few layer
2. Mid-level Feature: middle layer.
3. High-level Feature.

CNN

Neural Network Design, is a computational model consisting of interconnected neurons.

1. width & Depth of Neural Network

$4 \rightarrow 8 \rightarrow 6 \rightarrow 2$

b) Depth of a Neural Network

input $\rightarrow$ hidden $\rightarrow$ output

Model capacity:

Model capacity refers to the ability of a neural network to learn & represent complex patterns & functions from data.

Comparing Model capacity

Network A $5 \rightarrow 4 \rightarrow 2$

Network B $5 \rightarrow 20 \rightarrow 15 \rightarrow 10 \rightarrow 2$

Types of Neural Network

a) Feed Forward Neural Network (FNN)

- * Information flows in one direction
- * No feedback loop.

c) RNN

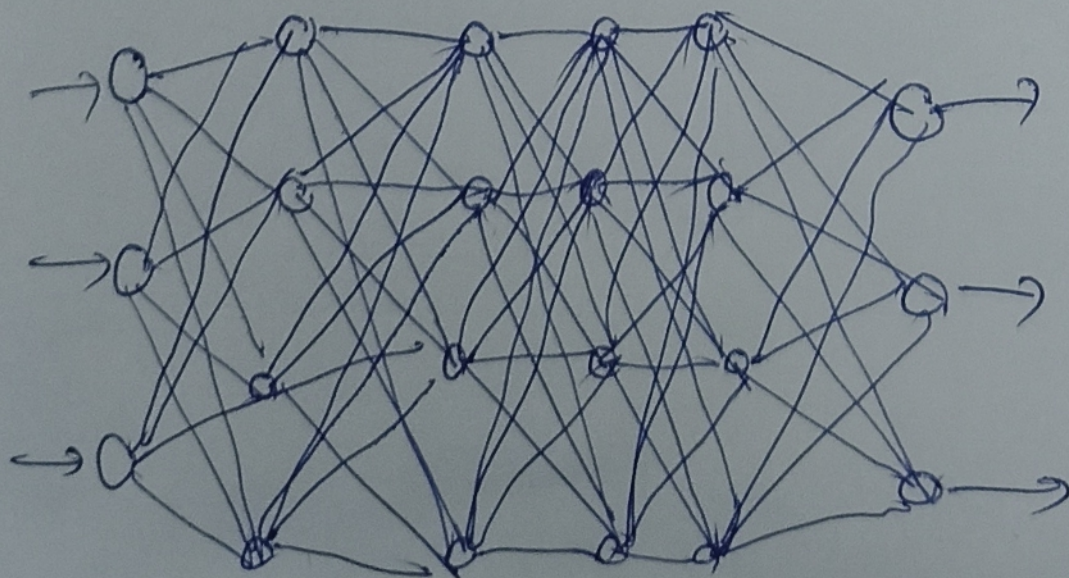
Processes sequential data.

- * maintains memory of previous points.

d) DNN

b) CNN

1. Input layer
2. convolutional layer
3. Activation layer
4. pooling layer.



28/7/23